The ETL process

Extract

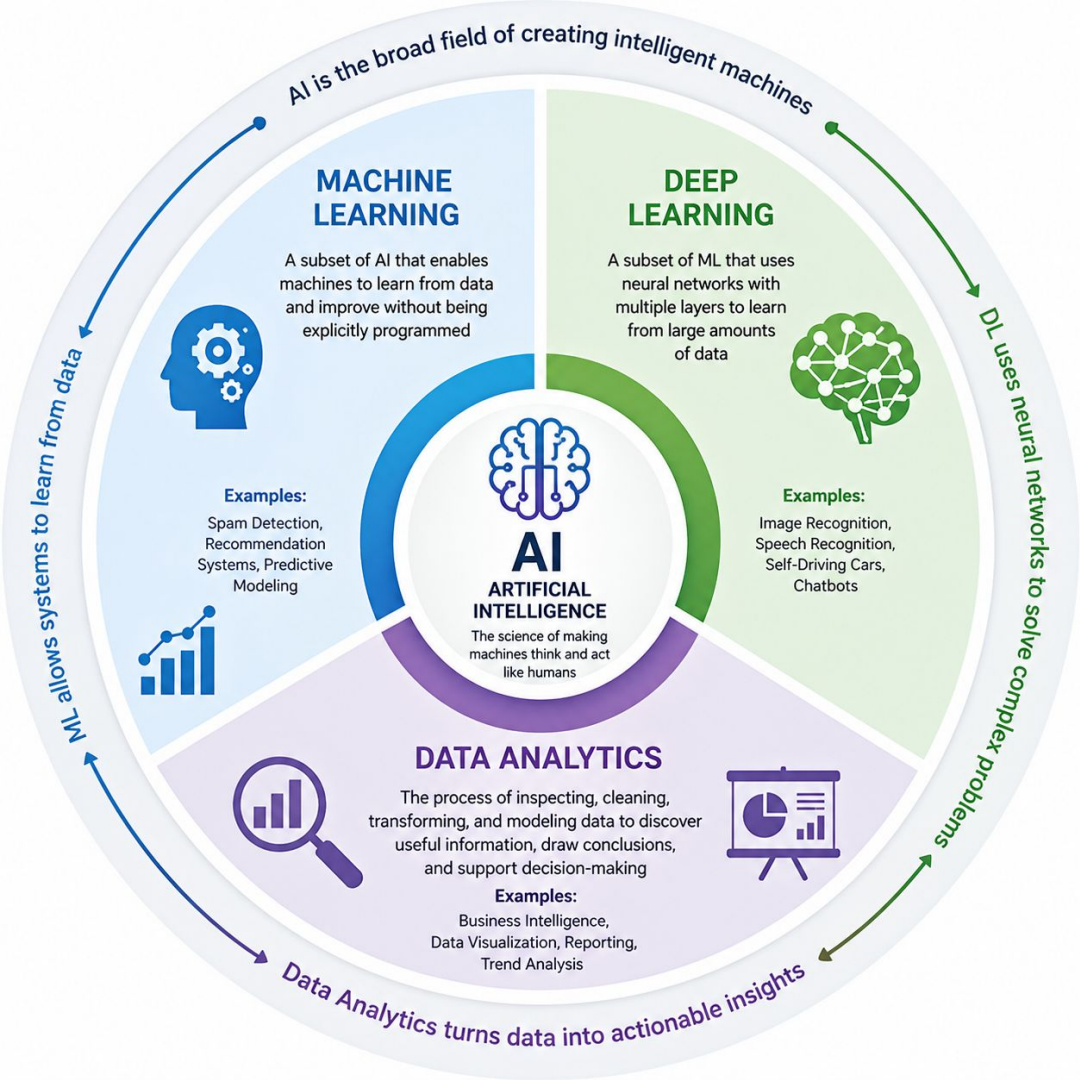
Retrieves and verifies data from various sources

Transform

Processes and organizes extracted data so it is usable

Load

Moves transformed data to a data repository


AI is the broad field of creating intelligent machines
MACHINE LEARNING
A subset of AI that enables machines to learn from data and improve without being explicitly programmed
Examples:
Spam Detection, Recommendation Systems, Predictive Modeling
DEEP LEARNING
A subset of ML that uses neural networks with multiple layers to learn from large amounts of data
Examples:
Image Recognition, Speech Recognition, Self-Driving Cars, Chatbots
AI
ARTIFICIAL INTELLIGENCE
The science of making machines think and act like humans
DATA ANALYTICS
The process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making
Examples:
Business Intelligence, Data Visualization, Reporting, Trend Analysis
ML allows systems to learn from data
DL uses neural networks to solve complex problems
Data Analytics turns data into actionable insights

Daily Global Activity

Category	Estimated Daily Volume
Emails Sent	Over 370 billion
Search Queries	~9 billion searches
Social Media Posts	~500 million tweets/posts
Video Content	~1 billion hours watched on YouTube
WhatsApp Messages	Over 120 billion messages

402.74 million TB => 463,000,000 TB

Predictive Analysis

“Predict, Perform, Prosper.”

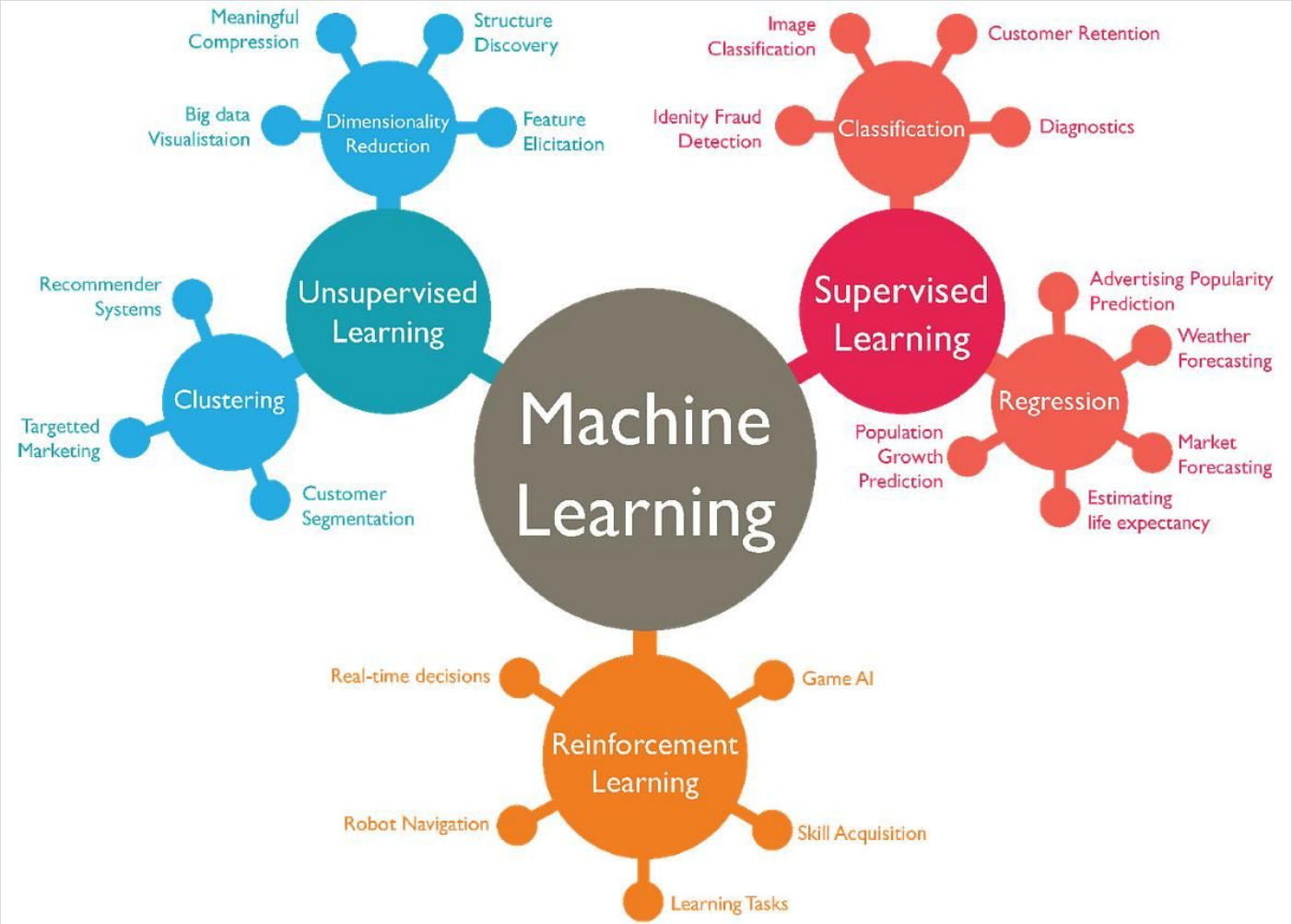
Data Analysis

Data analysis is the process of inspecting, cleaning, and interpreting data to extract useful information, draw conclusions, and support decision-making.

- Data analysis helps turn raw data into meaningful insights.

Types of Data Analysis

1. **Descriptive Analysis** → “What happened?”
2. **Diagnostic Analysis** → “Why did it happen?”
3. **Predictive Analysis** → “What is likely to happen next?”
4. **Prescriptive Analysis** → “What should we do about it?”


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Introduction to Scikit-Learn (sklearn)

scikit-learn: machine learning in Python — scikit-learn 1.8.0 documentation

The Scikit-Learn Philosophy

Scikit-Learn is built around the **Estimator API**. Whether you are predicting house prices or classifying flowers, the code pattern remains remarkably consistent:

1. **Instantiate**: `model = LinearRegression()`
2. **Fit**: `model.fit(X_train, y_train)`
3. **Predict**: `predictions = model.predict(X_test)`

Evaluation Metrics

1. Regression Metrics (Continuous Numbers)

- **Mean Absolute Error (MAE):** The average of all errors. If your MAE is 5.0, your predictions are, on average, 5 units off from the actual value.
- **R-Squared** (R^2): Also known as the "Coefficient of Determination." It tells us what percentage of the data's variation is captured by our model. A score of 1.0 is a perfect fit, while 0.0 means your model is no better than guessing the average.

Evaluation Metrics

2. Classification Metrics (Categories)

- **Accuracy:** The percentage of correct guesses. (

$$Accuracy = \frac{TP + TN}{Total}$$

). Note: Accuracy can be misleading if your classes are imbalanced!

Confusion Matrix: A N * N table that shows exactly where the model got "confused."

- **True Positives (TP)**: Correctly predicted positive.
- **False Positives (FP)**: Predicted positive, but was actually negative (Type I Error).
- **False Negatives (FN)**: Predicted negative, but was actually positive (Type II Error).

CONFUSION MATRIX

Actual \ Predicted	Predicted YES	Predicted NO
Actual YES	True Positive (TP) (Correctly predicted YES)	False Negative (FN) (Incorrectly predicted NO)
Actual NO	False Positive (FP) (Incorrectly predicted YES)	True Negative (TN) (Correctly predicted NO)

Thank You :)